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Introduction:

Taking Bearings: An Analytical Introduction to Artificial Intelligence in Knowledge Platforms and Open, Social Scholarship

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This research scan surveys an emergent and contested terrain: the interlaced developments in artificial intelligence, [[Generative Ai|generative AI]], [[Large Language Models|large language models]] (LLMs) and their predecessors, and the ways these systems are reshaping [[Knowledge Production|knowledge production]], circulation, and engagement across scholarly and public contexts. ^taking-bearings-an-analytical-introduction-p1

This scan uses artificial intelligence, or AI, as a broad umbrella term for computational systems designed to perform tasks associated with classification, prediction, reasoning, language processing, recommendation, perception, [[Automation|automation]], or decision support. It uses generative AI, or genAI, more specifically for systems that generate text, images, code, audio, video, or other outputs in response to prompts, queries, or other inputs. Large language models, or LLMs, are a prominent subset of contemporary genAI systems trained to model and generate language. Because public discourse often uses these terms interchangeably, this scan uses AI for the broader field and [[Infrastructure|infrastructure]], genAI where generative functions are specifically at issue, and LLM where the discussion concerns language-model systems in particular. The wider AI landscape includes technical architectures ranging from symbolic logic and “good old-fashioned AI” (GOFAI), through statistical [[Natural Language Processing|natural language processing]], to contemporary transformer-based genAI models, all situated within institutional and platform economies, cloud infrastructures, and evolving [[Governance|governance]] regimes ([[Jones 2023 - AI-in-History|Jones 2023]]; [[Anderson 2024 - AI-as-Philosophical-Ideology-A-Critical|Anderson 2024]]). These systems now recalibrate discretion, authority, and visibility, influencing what is recognized as knowledge, who produces it, and how it moves among diverse publics. ^taking-bearings-an-analytical-introduction-p2

A brief historical distinction is useful here. AI has not developed through a single technical or ideological pathway. Symbolic approaches, prominent in early AI, treated intelligence as the manipulation of explicit rules, formal representation, and logical structures. Connectionist approaches, by contrast, emphasized distributed pattern recognition across networks of weighted relations. Contemporary deep learning and transformer-based genAI inherit much more from the connectionist tradition, especially its reliance on large datasets, statistical association and learned representations rather than explicit rules. This inheritance helps explain why contemporary systems can be powerful aids for pattern detection, generation, and transformation, while remaining weak at grounding, verification, and accountable judgement. “Hallucinations”—false responses generated by AI that are especially misleading due to their plausible surfaces—stem from the inherent mathematical design of such systems, which are optimized to generate probable outputs rather than to verify claims within accountable epistemic communities. For fuller historical and critical accounts of these developments, see [[Jones 2023 - AI-in-History|Jones (2023)]], [[Syed Mustafa Ali 2023 - Histories-of-artificial-intelligence-a|Ali et al. (2023)]], [[Anderson 2024 - AI-as-Philosophical-Ideology-A-Critical|Anderson (2024)]], and [[Whiteley 2023 - Why-Artificial-Intelligence-Is-a-Misnome|Whiteley (2023)]]. ^taking-bearings-an-analytical-introduction-p3

What emerges from this mapping is a complex phenomenology of entanglement. Generative AI is woven in part from scholarship’s own materials—[[Open Access|open access]] literature, research data, [[Metadata|metadata]] schemas, [[Citation|citation]] networks, community-developed software, and collaborative data infrastructures. These are not the only, or even necessarily the largest, sources on which contemporary systems are trained. But they are unusually authoritative in that they help organize what counts as reliable knowledge, how claims are linked to evidence, how materials are discovered and how intellectual legitimacy is conferred. The AI systems that now analyze, summarize, and generate scholarly text therefore draw upon knowledge infrastructures whose value has been built over decades by researchers, editors, librarians, publishers, software communities, institutions, and publics. ^taking-bearings-an-analytical-introduction-p4

This recursive relationship positions scholarship as both one of AI’s important knowledge substrates and one of its objects of transformation, but the entanglement runs deeper still. Scholarship does not merely feed AI systems; it is increasingly mediated by them. Researchers discover literature through AI-powered recommendation engines, draft manuscripts with generative assistants, and navigate [[Peer Review|peer review]] processes augmented by algorithmic triage. Students encounter AI both as subject of study and as tool for learning, while institutions struggle to develop governance frameworks adequate to technologies that evolve faster than policy cycles. The recursive loop—scholarship training AI training scholarship—creates conditions where distinguishing human from machine contribution becomes progressively more difficult, and where the very categories through which we understand authorship, originality, and expertise require renegotiation. ^taking-bearings-an-analytical-introduction-p5

The organizing logic of this collection moves from foundational grounding through INKE-aligned focal points that structure the annotated bibliography itself. It first establishes essential contexts: the histories and theories of AI, including critical perspectives from Science and Technology Studies (STS) and Critical [[Digital Humanities]]; accounts of knowledge as relational, plural, and platformed; and the principles of Open Social Scholarship (OSS) as articulated by the Implementing New Knowledge Environments (INKE) Partnership and the Electronic Textual Cultures Lab (ETCL). It then follows three focal threads—“open,” “social,” and “scholarship”—that mark key sites where AI’s effects both align with and challenge OSS commitments to [[Accessibility|accessibility]], reciprocity, participatory governance, and public engagement ([[Arbuckle 2022 - An-Open-Social-Scholarship-Path-for-the|Arbuckle et al. 2022]]; [[El Khatib 2019 - Open-Social-Scholarship-Annotated-Biblio|El Khatib et al. 2019]]). ^taking-bearings-an-analytical-introduction-p6

Throughout, the ethos of Open Social Scholarship provides a normative coordinate: a set of principles for interpreting how AI mediates relations among researchers, institutions, communities, and publics. The focus is on how practices, infrastructures, and relations are changing, and how these shifts intersect with OSS values. Which concepts, practices, actors, infrastructures, and frictions are currently reconfiguring the conditions of knowledge and engagement? Where do these movements align with, complicate, or unsettle OSS principles of accessibility, reciprocity, and public value? By attending closely to these dynamics—openness entangled with [[Enclosure|enclosure]], [[Provenance|provenance]] linked to [[Trust|trust]], [[Bias|bias]] in tension with [[Diversity|diversity]], governance negotiated through [[Participation|participation]]—a conceptual scaffolding emerges for further analysis and empirical inquiry. ^taking-bearings-an-analytical-introduction-p7

A critical OSS response to AI need not be only defensive. The same systems that threaten enclosure, decontextualization, and epistemic flattening may, under different governance conditions, support broader forms of scholarly engagement: multilingual and plain-language summaries, alternative format outputs, assistive reading and navigation, metadata enrichment, participatory annotation, community translation, and more accessible pathways into specialized research. These possibilities in no way cancel the risks identified throughout this scan. Rather, they clarify the architectural task lying ahead for open, social scholarship: to ask what forms of AI-mediated scholarship can be built under conditions of reciprocity, [[Transparency|transparency]], care, public value and shared authority. ^taking-bearings-an-analytical-introduction-p8

Four Cross-Cutting Tensions

The sources assembled here reveal four cross-cutting tensions that structure AI’s engagement with scholarship. None admits of simple resolution—each marks a site where choices about AI design and deployment carry real stakes for knowledge production, and where different communities are actively negotiating trade-offs that no technical fix can settle. ^four-cross-cutting-tensions-p1

[[Operational Assistance vs Epistemic Authority|Operational assistance versus epistemic authority]]. While AI can reduce friction in many workflows, it raises fundamental questions about where judgement, [[Accountability|accountability]], and interpretive responsibility should reside, and how those boundaries should be renegotiated as tools, practices, and disciplinary norms change. AI excels at standardized tasks—formatting, initial drafts, pattern identification, literature triage—while struggling with the evaluative judgement that gives scholarship its value. The challenge lies in leveraging assistance without ceding authority, using AI capabilities while preserving human responsibility for consequential decisions. When does operational support cross into epistemic territory? How should that boundary be renegotiated as capabilities shift, practices stabilize, and disciplinary communities develop new norms? ^four-cross-cutting-tensions-p2

This boundary is historically and institutionally variable. Reference verification, metadata enrichment, formatting and duplication checks may increasingly be treated as operational tasks. Literature triage, evidence synthesis, translation and review assistance venture into more contested terrain. Assessments of originality, significance, methodological adequacy, interpretive force, and public consequence continue to require accountable human judgement. The problem of governance is therefore not one of simply drawing a fixed, indelible line once, but to define processes through which scholarly communities can revisit that line over time. ^four-cross-cutting-tensions-p3

The fluency of AI-generated text can mask the absence of understanding; the appearance of authority can obscure the statistical operations underneath. [[Sun 2025 - Large-Language-Models-in-Peer-Review-Ch|Sun (2025)]] emphasizes that models, by definition, are constrained in how they evaluate the novelty of research—they can recognize patterns but cannot evaluate whether patterns matter. This distinction is integral to peer review, research methods, pedagogy, and governance, and each requires clarity about where AI can appropriately assist and where human judgement must remain central. ^four-cross-cutting-tensions-p4

[[Openness vs Enclosure|Openness versus enclosure]]. The open infrastructure that has supported equitable access to knowledge is being absorbed into commercial AI systems in ways that may foreclose openness rather than extend it, and the trajectory of this tension will shape whether AI serves broad publics or narrow interests. Open access literature feeds model pretraining: This raises serious concerns about a new enclosure dynamic, in which freely accessible works become inputs for proprietary AI systems that generate commercial value without meaningful reciprocity. [[Pooley 2024 - Large-Language-Publishing-The-Scholarly|Pooley (2024)]] reveals a troubling symmetry in the underlying asymmetry: both commercial publishers and AI developers extract value from the scholarly [[Commons|commons]] while returning little to the researchers who create it. The licensing frameworks developed for human readers require substantial adaptation to govern machine consumption, where attribution and context preservation present fundamentally different challenges. Community-sourced data and [[Labour|labour]] underwrite AI functions, yet the question of whether benefits return to those communities remains contested. ^four-cross-cutting-tensions-p5

[[Technical Capability vs Organizational Capacity|Technical capability versus organizational capacity]]. The pace of AI development consistently outstrips the ability of institutions to evaluate, govern, and respond thoughtfully, and the gap between what AI can technically accomplish and what institutions can responsibly deploy constitutes a critical site of intervention. Building capacity for thoughtful AI adoption requires sustained investment in skills, governance structures, and evaluative frameworks. [[Weber 2023 - Organizational-Capabilities-for-AI-Imple|Weber et al. (2023)]] identify organizational capabilities as crucial mediators, noting failures often stem from organizational incapacity to manage sociotechnical complexity. Resource-intensive approaches assume capacities—technical staff, governance infrastructure, training budgets, and time for implementation—that are unevenly distributed across institutions and communities. These differences do not map neatly onto institutional size or prestige: large research universities may have more formal infrastructure, but also larger user bases, more complex bureaucracies, and interventions that can become less responsive to local needs. Smaller institutions such as community colleges, public libraries, independent scholarly organizations, and grassroots knowledge projects may face different constraints, including limited staffing, funding, or access to specialized expertise. Across all contexts, institutions are encountering unforeseen complexities for which existing governance structures may be poorly prepared. This tension requires attention to differential institutional positions and the development of governance models that remain humane, adaptable, and scalable across varied contexts. ^four-cross-cutting-tensions-p6

[[Efficiency vs Process|Efficiency versus process]]. Although AI accelerates tasks, scholarly knowledge earns its credibility through deliberate, iterative processes of review and community validation that efficiency-oriented tools may erode. Pedagogical and scholarly transformations require valuing cognitive work that AI cannot replicate, and education involves developing capacities through practice, not just acquiring outputs. The speed and fluency of AI-generated text can mask the erasure of the deliberative processes—reading, synthesizing, drafting, revising—through which understanding develops. [[Deng 2025 - Does-ChatGPT-Enhance-Student-Learning-A|Deng et al. (2025)]] find that ChatGPT improves student performance while reducing mental effort, raising questions about what learning means when cognitive work is offloaded to machines. The compression of research, writing, and discovery processes may generate outputs while eroding the developmental experiences that scholarly communities value. Process-centered assessment, transparent documentation, and attention to the formative dimensions of intellectual work become crucial responses. ^four-cross-cutting-tensions-p7

These tensions recur throughout the bibliography's domains. They structure debates about AI in peer review, pedagogy, platform governance, and infrastructure design. Understanding them as ongoing negotiations rather than problems awaiting solution helps position responsible engagement with AI's transformative effects. ^four-cross-cutting-tensions-p8

Fundamentals and Ideologies of AI

The prominence of generative AI and LLMs is rooted in layered histories. Early symbolic AI attempted to encode knowledge as explicit rules and representations. Statistical natural language processing redirected attention to learning patterns from data. Deep learning, accelerated by transformer architectures and large-scale computation, now underpins models that generate fluent text, code, and images ([[Jones 2023 - AI-in-History|Jones 2023]]). Each wave is embedded in institutional settings, funding regimes, and sociotechnical imaginaries that combine benchmark cultures, optimization logics, data abundance, and investor narratives of efficiency and disruption. The concept of "sociotechnical imaginaries" provides crucial analytical purchase for understanding how collective visions shape governance and discourse; as [[Richter 2025 - Imaginaries-of-Artificial-Intelligence|Richter, Katzenbach, and Schäfer (2025)]] demonstrate, these imaginaries function performatively, actively mobilizing resources, legitimating interventions, and establishing trajectories for development. The figure of "intelligence" functions here as a sociotechnical imaginary: a collective vision that orients research agendas, policy priorities, and public expectations, often sidelining questions of transparency, accountability, and epistemic pluralism. ^fundamentals-and-ideologies-of-ai-p1

Understanding contemporary AI requires genealogical attention to its intellectual origins and the sociotechnical imaginaries that have shaped its development. The transition from symbolic AI to statistical learning—from explicit knowledge representation to pattern recognition across massive datasets—represents a fundamental reorientation of assumptions about what counts as intelligence and whose knowledge gets represented. As [[Jones 2023 - AI-in-History|Jones (2023)]] argues, the shift toward "empiricist" approaches that capitalize on the "unreasonable effectiveness of data" threatens professions centred on particularity while demonstrating that AI functions as a reflection of human creations including their best and worst traits. ^fundamentals-and-ideologies-of-ai-p2

[[Anderson 2024 - AI-as-Philosophical-Ideology-A-Critical|Anderson's (2024)]] examination of John McCarthy's foundational program reveals AI as philosophical ideology—a vision of perfectly rational abstraction built upon objectivism and scientism that has "come to serve as a blind for increasing techno-corporate control of society." McCarthy's conception of artificial intelligence as a "stripped down model of human mental engagement without emotional qualities" established patterns that continue to shape the field: the subordination of interpretive judgement to formal optimization, the bracketing of contextual understanding in favor of scalable pattern matching, and the assumption that intelligence admits of decomposition into modular, transferable components. [[Syed Mustafa Ali 2023 - Histories-of-artificial-intelligence-a|Ali et al. (2023)]] extend this analysis by situating AI within broader genealogies of industrialization and colonialism, identifying threads of hidden labour, encoded behaviour, and cognitive injustice running through AI's history—positioning AI as the flagship of the Information Age conditioned by the Management Age. ^fundamentals-and-ideologies-of-ai-p3

The term "artificial intelligence" itself warrants critical scrutiny. As [[Whiteley 2023 - Why-Artificial-Intelligence-Is-a-Misnome|Whiteley (2023)]] observes, the phrase anthropomorphizes what is essentially computer-assisted statistical analysis, encouraging expectations and trust that the technology cannot warrant. This terminological slippage—from statistical prediction to "intelligence"—exemplifies how language shapes perception and, through perception, governance. The sources in this bibliography provide conceptual resources for more precise description that preserves space for human agency, critical [[Evaluation|evaluation]], and alternative design pathways. ^fundamentals-and-ideologies-of-ai-p4

Critical perspectives from Digital Humanities (DH) and Science, Technology and Society (STS) [KD1] supply key interpretive tools. They foreground the ways datasets, labels, and evaluation metrics encode values and worldviews; how training regimes replicate archival absences and classification politics; and how the defaults embedded in language, genre, and culture travel into outputs, with particular consequences for marginalized communities. [[Bender 2021 - On-the-Dangers-of-Stochastic-Parrots-Ca|Bender et al. (2021)]] critically analyze how larger models reproduce hegemonic views: size does not guarantee diversity, because the communities most underrepresented in [[Training Data|training data]] are also those most likely to bear the downstream costs of systems built without them. Automation theory highlights AI as a redistribution of discretion and authority: tasks that once required situated human judgement are delegated to systems that lack context, while accountability structures struggle to keep pace. ^fundamentals-and-ideologies-of-ai-p5

[[Berry 2025 - Synthetic-Media-and-Computational-Capita|Berry (2025)]] theorizes this moment as "the Inversion" under "computational capitalism" where machine-generated works reshape meaning-making infrastructure, producing "AI slop" that infiltrates information ecosystems with garbled meaning. His concept of "diffusionisation" describes how AI systems dissolve cultural forms into probabilistic vector spaces and reconstitute them through latent space manipulation—a process that transforms the infrastructure of meaning-making itself. This moves beyond mechanical reproduction toward what Berry theorizes as "post-consciousness," where boundaries between individual and synthetic consciousness become porous. The concept challenges scholars to attend not merely to AI outputs but to how generative systems transform the very grounds upon which experience and meaning are constituted. ^fundamentals-and-ideologies-of-ai-p6

Environmental and labour analyses add further layers, tracing the energy-intensive training runs and precarious annotation and moderation work that underpin current capabilities. [[Klein 2025 - Provocations-from-the-Humanities-for-Gen|Klein et al. (2025)]] argue that AI's statistical enactment of ideology imposes an objectivist European modernist framework upon data, making issues of "bias" unsolvable through technical means alone—even the distinction between "pure" and "toxic" training data reflects ideological commitments. They propose that smaller, purpose-built models developed through genuine interdisciplinary collaboration offer a path forward, one where "technical researchers must recognize the institutional asymmetries" that currently disadvantage humanistic approaches. ^fundamentals-and-ideologies-of-ai-p7

Relations with OSS-aligned communities complicate this picture. Open-source software, open datasets, and commons-based peer production have been central to AI's development; influential models rely on openly shared corpora and community-built tools. At the present inflection point, however, pressures toward enclosure are intensifying. Recent work by the [[Open Source]] Initiative on open source AI, including its 1.0 Open Source AI Definition, clarifies that releasing model weights alone is not equivalent to full openness. Meaningful openness also depends on access to documentation, code, data information, use rights, and other artifacts needed to understand, modify, and reuse a system. This clarification responds to broader concerns about “openwashing,” in which companies describe AI systems as open while releasing models under restrictive licenses, with incomplete documentation, or without access to key development artifacts. [[White 2024 - The-Model-Openness-Framework-Promoting|White et al. (2024)]] similarly develop the Model Openness Framework, establishing a three-tiered classification that shows AI “openness” exists on a spectrum, from minimal weight release to fuller disclosure of training data, intermediate checkpoints, code, documentation, and development practices. Rather than a simple binary between open and closed, the field is experimenting with hybrid forms—open weights under usage constraints, community checkpoints, and permissive yet bounded licenses—that seek to reconcile accessibility with risk management and [[Intellectual Property|intellectual property]] considerations. ^fundamentals-and-ideologies-of-ai-p8

[[Vake 2025 - Is-Open-Source-the-Future-of-AI-A-Data|Vake et al. (2025)]] find the open-source AI community depends on businesses to develop base models—there are few incentives for companies to release openly because doing so risks intellectual property and competitive advantage. This structural dependence raises questions about whether open AI can genuinely serve public interests or will remain subordinated to commercial imperatives. These choices shape not only technical access, but also epistemic legitimacy and the capacity for public oversight, critique, and repair. ^fundamentals-and-ideologies-of-ai-p9

Geographic Diversity in AI Imaginaries

The imaginaries circulating around AI vary significantly across geographic and political-economic contexts, challenging narratives of technological inevitability that treat AI development as following a single universal trajectory. [[Richter 2025 - Imaginaries-of-Artificial-Intelligence|Richter, Katzenbach, and Schäfer’s (2025)]] comparative analysis across the United States, China, and Germany reveals heterogeneous and often contradictory imaginaries even within single regulatory regimes. Where US discourse fragments across multiple geographic AI hubs, German imaginaries center on EU policy compliance and regulatory frameworks, while Chinese articulations align tightly with party-state directives that minimize local variation. ^geographic-diversity-in-ai-imaginaries-p1

Building on the varied sociotechnical imaginaries discussed earlier, [[Chung 2025 - Betting-on-UnCertain-Futures-Sociotec|Chung (2025)]] extends this analysis to economically advanced but geographically non-dominant Asian societies—Singapore, Hong Kong, and Taiwan—demonstrating how “techno-developmentalism” produces distinct imaginaries shaped by local political-economic conditions and historical trajectories. These varied articulations challenge the Global North/South binary that structures much [[Ai Governance|AI governance]] literature, revealing that AI development trajectories remain contingent and shaped by specific political choices rather than following universal technological imperatives. ^geographic-diversity-in-ai-imaginaries-p2

The imaginaries circulating around AI shape technical design itself. Van Es and Nguyen (2024) demonstrate how ChatGPT’s self-representations—82% depicting humanoid forms—strategically cultivate anthropomorphic imaginaries that encourage overlooking ethical and legal challenges. The friendly assistant persona, surrounded by books and radiating futuristic light, constructs a humanoid image of trustworthy intellectual companionship that obscures the statistical operations underneath. ^geographic-diversity-in-ai-imaginaries-p3

For scholarly communities, these developments necessitate specifically humanities-oriented engagement. [[Chun 2023 - The-Crisis-of-Artificial-Intelligence-A|Chun and Elkins (2023)]] argue that humanities-oriented pedagogy offers distinctive resources for this moment, insisting that the most effective AI tools amplify rather than replace human intellectual engagement—a claim developed more fully in the Critical Literacies section below. ^geographic-diversity-in-ai-imaginaries-p4

Knowledge Foundations

In this framework, knowledge appears as a relational practice rooted in communities, languages, and infrastructures. Three dimensions are especially salient for understanding AI's impact: diversity, mobilization, and platforms. ^knowledge-foundations-p1

Diversity. Epistemic, cultural, linguistic, and methodological plurality are conditions for robust, public-facing scholarship. Models trained predominantly on dominant corpora risk misrecognition and erasure, flattening the diversity that constitutes human knowledge. [[Arbuckle 2020 - How-Can-We-Broaden-and-Diversify-Humanit|Arbuckle (2020)]] articulates diversity as fundamental to legitimate knowledge translation, emphasizing inclusive modes of engagement and participatory approaches that honor varied ways of knowing. In response, approaches such as community governance of data, participatory evaluation, and multilingual modeling work to adapt tools to difference rather than requiring difference to conform to tools. [[Crompton 2020 - Familiar-Wikidata-The-Case-for-Building|Crompton et al. (2020)]] address how community governance can safeguard plural representation in data infrastructures like Wikidata, demonstrating that technical systems encode choices about whose knowledge counts and how it gets categorized. The [[Fair Principles|FAIR principles]]—Findable, Accessible, Interoperable, Reusable—offer a technical framework for data stewardship, while the CARE principles—Collective Benefit, Authority to Control, Responsibility, Ethics—introduce dimensions of sovereignty, [[Consent|consent]], and community benefit that are especially important for Indigenous data and other contexts shaped by power asymmetries ([[Carroll 2020 - The-CARE-Principles-for-Indigenous-Data|Carroll et al. 2020]]; [[Carroll 2021 - Operationalizing-the-CARE-and-FAIR-Princ|Carroll et al. 2021]]). ^knowledge-foundations-p2

Mobilization. Knowledge mobilization concerns how research travels and is taken up across domains: from scholarship to policy, from specialist discourses to public understanding, from the classroom to community settings. Generative tools can accelerate synthesis and enhance accessibility by producing multilingual summaries, plain-language translations, and alternative-format outputs. The same tools can also increase the volume and velocity of decontextualized or misleading claims. [[Winter 2023 - Introduction-Open-Scholarship-Policy-in|Winter (2023)]] frames policy as a vehicle for mobilization emphasizing provenance and accountability—mechanisms that become ever more crucial when automated systems mediate between research and its publics. [[Jensen 2023 - Introduction-Digital-Knowledge-Commons|Jensen (2023)]] connects commons-based infrastructures to mobilization pathways, highlighting how [[Interoperability|interoperability]] and community practices shape whether knowledge flows toward public benefit or private capture. A central challenge is ensuring that speed does not erode fidelity, and that reach does not come at the expense of context or community consent. ^knowledge-foundations-p3

The Canadian HSS Commons exemplifies mobilization infrastructure designed around principles of reciprocity and community responsiveness. [[Winter 2020 - Foundations-for-the-Canadian-HSS-Commons|Winter et al. (2020)]] describe how this platform instantiates governance mechanisms, provenance tracking, and dialogic engagement in a Canadian context, offering a concrete model for how digital research communities might align with OSS values. The Commons demonstrates that alternatives to extractive platform models are possible when design priorities center community benefit and participatory governance. ^knowledge-foundations-p4

Infrastructures and Platforms. Repositories, journals, preprint servers, and social scholarly tools form the infrastructural backbone of [[Scholarly Communication|scholarly communication]] due to their ubiquity, reliability and invisibility (Plantin et al. 2018). AI-mediated functions such as search, recommendation, summarization, moderation, and peer review reshape relations of production, discoverability, and legitimacy. Turning research outputs into training data for AI technologies creates a process of platformization of digital research infrastructure, in which the research process is commodified and turned into data for publishers and corporations to exploit while excluding the research community from the governance of these infrastructures. [[Bullard 2023 - Describing-the-HSS-Commons-The-View-fro|Bullard (2023)]] reveals how metadata shapes these dynamics, demonstrating that descriptive schemas determine what gets found, how it gets categorized, and whose work becomes visible. [[Goddard 2021 - Persistent-Identifiers-as-Open-Research|Goddard (2021)]] positions persistent identifiers as crucial connective tissue for provenance and interoperability, enabling the traceability that accountability requires. Whether platforms act as commons or proprietary funnels depends on decisions about interoperability, open standards, licensing, and governance. [[Colbert-Lewis 2024 - The-Citation-Economy-as-a-Site-of-Extrac|Colbert-Lewis et al. (2024)]] examine the citation economy as a site of extraction for [[Surveillance|surveillance]] publishing, documenting how publishers have transitioned to technology-driven data brokers that monetize researcher behavior alongside research outputs. Seemingly technical or back-end decisions structure how knowledge circulates and who benefits from that circulation. ^knowledge-foundations-p5

Open Social Scholarship Principles

Open Social Scholarship, as developed by INKE and ETCL, characterizes scholarship as a public, collaborative practice sustained by open, participatory, and ethically governed infrastructures ([[Arbuckle 2022 - An-Open-Social-Scholarship-Path-for-the|Arbuckle et al. 2022]]; [[El Khatib 2019 - Open-Social-Scholarship-Annotated-Biblio|El Khatib et al. 2019]]). Within this frame, AI is evaluated not solely on technical metrics but on whether it strengthens dialogic exchange, lowers barriers without reproducing harm, and supports reciprocal engagement rather than unilateral extraction. Openness is coupled with care; accessibility with accountability; participation with shared authority. [[Arbuckle 2019 - Modelling-Open-Social-Scholarship-Within|Arbuckle and Maxwell (2019)]] move beyond open access by emphasizing infrastructure choices and governance as sites where values are enacted—recognizing that technical decisions embed normative commitments about who can participate, on what terms, and toward what ends. [[Maxwell 2015 - Beyond-Open-Access-to-Open-Publication-a|Maxwell (2015)]] argues that “access” alone proves insufficient; usability and community-responsive workflows reposition openness as an ecosystem with embedded care and accountability. ^open-social-scholarship-principles-p1

This also gives OSS a constructive role in relation to AI. Rather than approaching AI only as an external threat to be resisted, OSS can help specify what accountable alternatives would require: community-shaped tools, transparent provenance, accessible interfaces, multilingual and multimodal engagement, benefit-sharing where public or community knowledge is used, and governance arrangements that keep interpretive authority distributed rather than concentrated in proprietary systems. ^open-social-scholarship-principles-p2

Several principles translate into analytical vantage points for observing AI’s effects:

Openness with care balances access with consent and community control, recognizing that uncontrolled circulation can facilitate extraction when reciprocity is absent ([[Carroll 2020 - The-CARE-Principles-for-Indigenous-Data|Carroll et al. 2020]]). Some knowledge requires protection; openness can become extractive when divorced from attention to power asymmetries and community interests. ^open-social-scholarship-principles-p3

Participation with shared authority embeds publics and communities in the design, evaluation, and governance of AI tools, rather than positioning them as passive recipients. Engagement becomes genuine partnership, not consultation that leaves final decisions to technical experts. ^open-social-scholarship-principles-p4

Accessibility is treated as multidimensional, spanning language, different ableness, technical literacy, and the digital divides that structure who can engage with AI-mediated scholarship. True accessibility requires attention to the multiple barriers—linguistic, economic, infrastructural—that shape unequal participation. ^open-social-scholarship-principles-p5

Reciprocity and benefit-sharing acquire particular weight where community-sourced data, scholarship, cultural material or labour underwrite AI functions; the question of whether benefits return to those communities becomes central. In heterogeneous training-data environments, reciprocity cannot always mean simple one-to-one compensation or attribution. It may also require collective mechanisms: investment in public infrastructure, support for community-governed datasets, licensing arrangements, provenance standards, access rights, audit mechanisms, and benefit-sharing models that recognize the distributed labour through which knowledge systems are built. ^open-social-scholarship-principles-p6

Infrastructural responsibility assigns institutions ongoing obligations to steward provenance, auditability, and accountability across the AI lifecycle—from adoption through monitoring, incident response, and iterative refinement. ^open-social-scholarship-principles-p7

These principles serve as lenses that help clarify where choices around AI align with OSS values and where they introduce new tensions. They support a descriptive, attuned stance that remains open to emerging research, changing practices, and the contested meanings that animate debates over AI's place in scholarship. ^open-social-scholarship-principles-p8

AI and "Open"

Openness operates across several registers—access, data, source, science, and social scholarship—and AI interacts differently with each. The recursive relationship between AI and openness demands particular attention: open access literature, [[Open Data|open data]], open source software, open standards, and community-maintained infrastructures have all contributed to the conditions under which contemporary AI systems became possible. Yet these resources now circulate within systems that often tend toward enclosure, secrecy, and commercial capture. The result is not a simple opposition between open and closed, but a contested field in which openness can serve public knowledge, enable scrutiny, and support community adaptation, while also becoming a reservoir from which private systems extract value without adequate attribution, consent, or reciprocity. ^ai-and-open-p1

Open Access

Open access literature forms one important input into model pretraining, retrieval-augmented generation, summarization, and later synthesis, while a dark shadow has fallen over how works licensed for broad human access are being repurposed for machine processing, private capture, and commercial value extraction. [[Pooley 2024 - Large-Language-Publishing-The-Scholarly|Pooley (2024)]] critically examines how commercial publishers frame AI companies as "free-riders" while themselves profiting from unpaid scholarly labour, extending "surveillance publishing" into AI domains. His analysis reveals a troubling symmetry: both commercial publishers and AI developers extract value from the scholarly commons while returning little to the researchers who create it. [[Kember 2023 - The-Corporate-Capture-of-Open-Access-Pub|Kember and Brand (2023)]] document how OA policies may unintentionally contribute to greater consolidation enabling "corporate capture"—a warning that good intentions around openness can produce perverse outcomes when power asymmetries remain unaddressed. ^open-access-p1

Licensing frameworks and machine-access provisions influence whether open access functions as a commons or a reservoir for extraction. [[Luth 2025 - The-Use-of-Wikipedia-Wikimedia-and-Ope|Luth

(2025)]] examines how stakeholders navigate AI training on [[Creative Commons]] content, finding compliance often uncertain and highlighting strategies for protecting shared knowledge while sustaining open ecosystems. [[Smith 2025 - Licensing-of-Text-for-Generative-AI-Lea|Smith (2025)]] analyzes how existing licenses perform in [[Machine Learning|machine learning]] contexts, noting most were drafted without anticipating large-scale AI training and proposing machine-readable licensing provisions that could clarify terms and enable technical enforcement. These analyses suggest that the licensing frameworks developed for human readers require substantial adaptation to govern machine consumption, where attribution and context preservation present fundamentally different challenges. ^open-access-p2

AI thus makes open licensing a live infrastructural question rather than a settled legal background condition. Licensors, libraries, funders, publishers, repositories, and scholarly communities will need to clarify how open licenses apply to machine reading, text and data mining, model training, retrieval-augmented generation, and downstream synthetic outputs. The important question here is whether the use of open materials preserves attribution, context, reciprocity and the public purposes for which openness was advanced. ^open-access-p3

The dual role of open access—as both input to AI development and public good in its own right—underscores the importance of provenance-aware summarization, appropriate attribution, and consent mechanisms that reflect authors' and communities' intentions. When AI models synthesize open access literature, questions of attribution become vexed: whose work is represented in a probabilistic blend? How can communities maintain oversight of how their contributions are used? ^open-access-p4

Open Data

Open data, crucial for [[Reproducibility|reproducibility]] and model improvement, raises issues of privacy, consent, and community sovereignty, especially in relation to sensitive or Indigenous data. The CARE principles complement FAIR by centering collective benefit, authority to control, responsibility, and ethics ([[Carroll 2020 - The-CARE-Principles-for-Indigenous-Data|Carroll et al. 2020]]). [[Carroll 2020 - The-CARE-Principles-for-Indigenous-Data|Carroll et al. (2020)]] ground these principles in the structural biases and colonization processes producing “epistemicide,” directly addressing tensions between Indigenous [[Data Sovereignty]] and AI's voracious data demands. Their framework demonstrates that open data governance constitutes a political practice, where broad data sharing and algorithmic extraction can undermine the rights, consent, and collective interests of communities whose knowledge becomes training material. ^open-data-p1

[[Carroll 2021 - Operationalizing-the-CARE-and-FAIR-Princ|Carroll et al. (2021)]] demonstrate FAIR and CARE operationalized together. Because many scientific datasets contain [[Indigenous Knowledge]], applying CARE principles ensures this data is oriented toward community wellbeing through rigorous provenance tracking and appropriate governance models. Arrangements such as data trusts, tiered access, and purpose-bound licenses illustrate how openness can be conditioned: open to whom, for which purposes, under what accountability terms? ^open-data-p2

Documentation practices emerge as crucial infrastructure for responsible AI development. [[Bender 2018 - Data-Statements-for-Natural-Language-Pro|Bender and Friedman (2018)]] propose data statements as critical enabling infrastructure to reduce exclusion and bias in language technologies, arguing that such documentation can prevent misrepresentation and support more ethical system development. [[Boyd 2021 - Datasheets-for-Datasets-Help-ML-Engineer|Boyd (2021)]] empirically demonstrates that documentation frameworks increase engineers' ethical sensitivity—through think-aloud sessions, practitioners who received Datasheets identified ethical issues earlier and more frequently. This finding challenges assumptions that openness automatically leads to fairer systems; structured documentation actively shapes how practitioners recognize and respond to potential harms. ^open-data-p3

Open Source

Open source dynamics in AI include open weights, permissive licenses, and community checkpoints that support verification, plural innovation, and public-interest development. At the same time, discussions of misuse and dual-use emphasize capability control and risk. [[White 2024 - The-Model-Openness-Framework-Promoting|White et al. (2024)]] establish a Model Openness Framework with three-tiered classification, codifying 17 components that expose the requirements for reproducibility and scrutiny. Graduated openness—incremental model releases, safety mitigations, and governance charters—appears as one path toward balancing transparency with harm reduction. The evolution of open models will influence the accessibility of AI to under-resourced institutions, public-interest researchers, and communities seeking to adapt technologies to local contexts. ^open-source-p1

The Open Source Initiative's Open Source AI Definition 1.0, released in 2024, provides criteria for describing an AI system as "Open Source AI," including the freedoms to use, study, modify, and share, access to the preferred form for modification, and sufficient information about training data provenance. As a definition rather than a license or implementation practice, it offers a community standard for assessing whether licenses, releases, and documentation practices support genuine openness, including whether "open weights" claims are matched by the artifacts, permissions, and transparency needed to avoid openwashing. ^open-source-p2

Open Science

Open science principles encourage reproducible workflows—transparent notebooks, containers, pipelines, and documentation—while AI introduces dependencies that can be opaque and difficult to reproduce. Practices such as open benchmarking, model cards, data cards, and documentation standards aim to make training data, evaluation metrics, and known limitations more visible. The push and pull between speed and scrutiny is pronounced in AI-assisted research: rapid exploration can generate insights, yet robust claims require validation, transparency, and often slower processes. ^open-science-p1

The tension between efficiency and process manifests acutely in open science contexts. AI can accelerate literature review, code generation, and data analysis, but the compression of these processes may obscure the iterative work through which understanding develops. Open science's emphasis on showing work—through preregistration, open notebooks, and transparent reporting—provides resources for maintaining visibility even as AI augments research workflows. ^open-science-p2

Open Social Scholarship

Within open social scholarship, AI is approached as a means to enhance public dialogue and shared inquiry. Examples include co-authorship with non-academic collaborators, explicit attribution of AI assistance, and participatory governance of tools in community-engaged projects ([[Arbuckle 2022 - An-Open-Social-Scholarship-Path-for-the|Arbuckle et al. 2022]]). Alignment with OSS principles is assessed by whether AI-supported practices sustain reciprocity, accessibility, and shared authority in knowledge production, and whether technological mediation amplifies rather than displaces human voice, agency, and accountability. ^open-social-scholarship-p1

The feedback relationship—scholarship as training data, AI as scholarly tool—manifests throughout these registers. Open access literature trains models that then summarize open access literature. Peer review norms shape training data that then assists peer review. Pedagogical materials become training data for tools used in teaching. This recursion creates feedback loops whose effects on scholarly practice remain to be fully understood. ^open-social-scholarship-p2

AI and “Social”

The “social” dimension concerns the distribution of attention, credibility, and participation within and beyond academia. AI now helps mediate content creation, [[Curation|curation]], and moderation, altering these distributions in ways that merit close scrutiny. ^ai-and-social-p1

Platforms and Synthetic Content

Platforms hosting synthetic content[KD2] must contend with provenance and identity. As generated text, images, and deepfake media proliferate, distinguishing among forms of authorship becomes more complex, with direct implications for trust and credibility. Van Es and Nguyen (2024) show that these anthropomorphic imaginaries are not incidental: they are actively constructed through ChatGPT’s own self-representations, which consistently project a friendly, humanoid intellectual companion that obscures the statistical operations underneath. [[Mitra 2024 - Sociotechnical-Implications-of-Generativ|Mitra, Cramer, and Gurevich (2024)]] diagnose “information access” under generative AI as “sociotechnical settlement”—models “intermediate credibility, reshape attention, and re-allocate epistemic authority.” Their framing positions AI-mediated search and synthesis as fundamentally reconfiguring who gets heard, whose claims become visible, and whose expertise counts. ^platforms-and-synthetic-content-p1

Recommendation systems juggle personalization, diversity, and serendipity, shaping canon formation and the visibility of marginalized voices. Attention economies are being reshaped: some work is surfaced and amplified; other work is obscured, sometimes by design and sometimes as a side effect of algorithmic optimization. Content credentials—cryptographic signatures and rich metadata—attempt to record origin and transformation history, while questions persist about whether such mechanisms can scale to match the velocity of synthetic content production. For scholarly communication, such mechanisms should be treated as part of the infrastructure of citation, preservation, disclosure and accountability; their value depends on whether they can travel across repositories, journals, platforms and public interfaces without being stripped of context. ^platforms-and-synthetic-content-p2

Governance, Moderation, and Policy

Leadership, governance, moderation, and policy become iterative and layered rather than singular and top-down. Universities, libraries, journals, and repositories intersect with platform governance in setting norms for AI-assisted authorship, disclosure, review, and engagement. [[Sartori 2022 - A-socio-technical-perspective-for-the-fut|Sartori and Theodorou (2022)]] view AI as a “magnifying glass” amplifying existing inequities—a framing that clarifies bias as an older problem than the technology and locates responsibility in social structures and institutional choices. Their analysis challenges approaches that treat fairness as achievable through algorithmic adjustment alone, arguing that human control requires institutional mechanisms for contestation that acknowledge AI practice as fundamentally political. ^governance-moderation-and-policy-p1

[[Akter 2021 - Algorithmic-bias-in-data-driven-innovati|Akter et al. (2021)]] theorize [[Algorithmic Bias|algorithmic bias]] as emerging from interacting sources—data bias, method bias, and societal bias—demonstrated through Australia’s Robo-Debt scheme, which inherited and amplified existing inequities through automated decision-making. Their framework implicitly contests technological determinism by revealing how managerial choices about data collection priorities, acceptable error rates, and stakeholder consultation shape algorithmic outcomes in ways that exceed engineering decisions. ^governance-moderation-and-policy-p2

[[Jensen 2022 - Fostering-Digital-Communities-of-Care-S|Jensen et al. (2022)]] address governance and moderation as platform design commitments aligned with OSS values, showing how choices about community safety and trust shape the kinds of scholarship platforms can support. Transparency reports, appeal procedures, and community-based moderation seek to address harms while preserving legitimate expression. Policy becomes experimental, subject to revision as impacts, unintended consequences, and community norms evolve. ^governance-moderation-and-policy-p3

Critical Literacies

Critical literacies, for both scholars and publics, extend beyond operational proficiency. They involve understanding model limits, recognizing sources of bias, interpreting provenance signals, and practicing verification. [[Chun 2023 - The-Crisis-of-Artificial-Intelligence-A|Chun and Elkins (2023)]] position AI's advancement as precipitating interlocking crises demanding pedagogical response grounded in reflective and collaborative meaning-making, explicitly connecting pedagogical design to civic preparation and positioning students as "critically engaged citizens capable of interrogating the social and economic implications of algorithmic mediation." [[Zeffiro 2024 - -EASST4S|Zeffiro (2024)]] reveals how corporate narratives of AI as "democratizing force" obscure differential vulnerabilities while "automating insecurities"—a concept capturing how AI tools institutionalize particular understandings of risk that reflect corporate priorities rather than community needs. ^critical-literacies-p1

These literacies include skills such as documenting AI use, distinguishing synthesis from source, reading expressions of uncertainty, and maintaining awareness of the gap between fluency and accuracy. The ease with which AI produces confident-sounding text makes discrimination skills more important: fluency does not indicate accuracy, and the appearance of authority can mask the absence of understanding. ^critical-literacies-p2

Global Asymmetries and Epistemic Centralization

AI's geopolitical dimensions draw attention to questions of influence, globalism, and colonization. Training regimes often reproduce global asymmetries in language and corpus composition: English-language and Global North sources dominate, while linguistic and epistemic diversity from other regions receive less representation. [[Bender 2021 - On-the-Dangers-of-Stochastic-Parrots-Ca|Bender et al. (2021)]] critically analyze how larger models reproduce hegemonic views, noting that the damages associated with large-scale language models—environmental costs, representational harms, concentration of power—are more likely to fall on marginalized populations. ^global-asymmetries-and-epistemic-centralization-p1

[[Chung 2025 - Betting-on-UnCertain-Futures-Sociotec|Chung (2025)]] extends geographic scope to Asia's advanced economies, revealing how techno-developmentalism produces distinct imaginaries shaped by local political-economic conditions. Multilingual models, community-owned datasets, and South–South collaborations seek to counteract epistemic centralization, yet the structural dynamics of extraction without reciprocity remain strong. [[Arthur 2021 - Open-Scholarship-in-Australia-A-Review|Arthur et al. (2021)]] reveal how national policy shapes whether openness reduces or reproduces inequities, showing that structural barriers—funding models, institutional incentives, infrastructure gaps—mediate between stated commitments to openness and their realization in practice. ^global-asymmetries-and-epistemic-centralization-p2

Debates over determinism, diversity, and justice challenge narratives of technological inevitability. [[Sartori 2022 - A-sociotechnical-perspective-for-the-fut|Sartori and Theodorou (2022)]] extend this analysis to the global scale, emphasizing that adoption choices, design priorities, and evaluation metrics remain matters of collective agency—and that the communities most affected by AI's asymmetries are precisely those least represented in the governance structures that shape them. Justice-oriented evaluation foregrounds the distribution of harms and benefits, rather than focusing solely on aggregate performance: who is helped, who is harmed, and along which axes of power and marginalization? ^global-asymmetries-and-epistemic-centralization-p3

Humanity, Community, Connection, and the Affective Dimensions of AI Mediation

Beyond the technical and governance dimensions, AI mediation carries affective and relational aspects that shape scholarly communication in subtle but significant ways. Generative systems influence the tone and velocity of discourse and can intensify polarization or foster connection depending on design choices and deployment contexts. The ability to flood channels with synthetic text or avatars strains existing trust infrastructures, creating new challenges for maintaining the relational fabric of scholarly communities. ^humanity-community-connection-and-the-p1

Design choices that emphasize care—thoughtful moderation, friction for high-risk actions, community control of spaces—become part of scholarly platform ethics. These choices counter acceleration and decontextualization that undermine deliberation and mutual recognition. The question moves beyond whether AI produces accurate outputs, but whether AI-mediated environments support the kinds of dialogue, disagreement, and collaborative inquiry that scholarship requires. Connection and community constitute values that technical efficiency can erode if left unattended. ^humanity-community-connection-and-the-p2

[[Jensen 2022 - Fostering-Digital-Communities-of-Care-S|Jensen et al. (2022)]] explore how digital communities of care can be fostered through attention to safety, security, and trust—values that require ongoing cultivation rather than one-time technical implementation. The relational dimensions of scholarship—mentorship, collaboration, peer recognition, collegial support—may be affected by AI mediation in ways that exceed questions of accuracy or efficiency. ^humanity-community-connection-and-the-p3

Environmental and Labour Costs

Material costs, both labour and environmental, ground AI capability in ways that often remain invisible. The AI pipeline depends on precarious annotation and moderation labour, often transnational and underpaid. Workers in the Global South perform essential tasks—labeling training data, flagging harmful content, refining model outputs—under conditions that rarely feature in celebratory narratives of AI advancement. This hidden labour connects AI development to broader patterns of extraction and externalization, where the costs of technological systems are displaced onto vulnerable populations. ^environmental-and-labour-costs-p1

Energy and water consumption for training and deployment contribute to climate impacts that intersect with environmental justice concerns. [[Bender 2021 - On-the-Dangers-of-Stochastic-Parrots-Ca|Bender et al. (2021)]] observe that the damages associated with large-scale language models—environmental costs, representational harms, concentration of power—“are more likely to fall on marginalized populations.” The communities least likely to benefit from AI capabilities bear disproportionate burdens from their development. ^environmental-and-labour-costs-p2

Disclosure norms, ethical procurement, and commitments to “green AI” bring these dimensions into infrastructural planning rather than treating them as externalities. For institutions committed to sustainability and justice, AI adoption cannot be evaluated solely on functional criteria; the material conditions of AI production become relevant to responsible deployment decisions. This connects AI governance to broader institutional commitments around environmental responsibility and labour ethics. ^environmental-and-labour-costs-p3

AI and “Scholarship”

Across the scholarly lifecycle, AI now participates in research, writing, teaching, review, and administration, changing how practice and infrastructure co-evolve. The tension between operational assistance and epistemic authority manifests at each stage. ^ai-and-scholarship-p1

Research Methods and Practices

Research methods and practices incorporate LLMs for literature review, coding, data cleaning, translation, and analysis. Key questions concern how assistance is documented, how outputs are validated, and how communities renegotiate the line between exploratory use, operational support, interpretive contribution, and settled scholarly claim. [[Mehlenbacher 2024 - Synthetic-Genres-Expert-Genres-Non-Spe|Mehlenbacher, Balbon, and Mehlenbacher (2024)]] examine "synthetic genres" that push apart definitions of "information" and "knowledge," suggesting that AI-generated text occupies a novel categorical space requiring new frameworks for evaluation. [[Bozkurt 2024 - GenAI-et-al-Cocreation-Authorship-Ow|Bozkurt (2024)]] addresses authorship implications, proposing the aiTARAS Framework while maintaining that responsibility ultimately belongs to the human author. Maintaining epistemic standards while exploring new efficiencies requires treating AI as a tool supporting thought and analysis, [KD3] not as a substitute for disciplinary judgement. ^research-methods-and-practices-p1

Domain-specific norms—interpretive rigor in the humanities, reproducibility in quantitative fields—must shape appropriate use. The question of when AI assistance becomes AI authorship remains contested, with implications for credit, accountability, and the integrity of scholarly claims. Documenting AI use transparently allows readers to assess the epistemic status of outputs and evaluate whether human judgement remains central to the knowledge claims of a given piece. ^research-methods-and-practices-p2

Scholarly Outputs

Forms of scholarly output are changing as generative tools support multimodal and interactive work: dynamic narratives, layered visualizations, and synthesis artefacts that draw together diverse sources. [[Bergstrom 2024 - A-Third-Transformation-Generative-AI-an|Bergstrom and Ruediger (2024)]] anticipate "discovery, interpretation, and writing services" becoming integrated suites while noting concerns about platform intermediation that could concentrate power over scholarly communication. Authorship, attribution, and preservation practices need to adapt: whose contributions are recognized, how AI assistance is disclosed, and how AI-mediated outputs are archived and made citable. Provenance and disclosure become part of the paratext, informing evaluations of reliability and situating works within ongoing conversations. ^scholarly-outputs-p1

Teaching and Pedagogy

Teaching and pedagogy integrate AI as both subject and instrument. [[Tan 2024 - Shaping-Integrity-Why-Generative-Artifi|Tan and Maravilla (2024)]] position AI as "catalyst necessitating pedagogical transformation—shifting from transmission models toward environments where students actively construct knowledge." This reframing treats AI as an occasion for rethinking educational purposes and methods, emphasizing the human capacities—critical judgement, ethical reflection, creative synthesis—that AI systems cannot replicate. ^teaching-and-pedagogy-p1

[[Deng 2025 - Does-ChatGPT-Enhance-Student-Learning-A|Deng et al. (2025)]] meta-analyze 69 studies finding ChatGPT improves performance while reducing mental effort, raising questions about what learning means when cognitive work is offloaded to machines. Their findings suggest that traditional assessment measures require recalibration to capture the thinking processes that educational institutions value. The tension between efficiency and process manifests acutely here: if AI can produce essays, what is being assessed? Process-centered assessment emphasizes transparency, ethical reflection, and the development of critical literacies that enable students to assess model outputs, recognize limitations, and engage thoughtfully with AI-mediated knowledge. ^teaching-and-pedagogy-p2

[[Lee 2025 - Prompt-Engineering-in-Higher-Education|Lee and Palmer (2025)]] establish prompt engineering as emergent practice mirroring "Socratic dialogue," suggesting continuities between AI interaction and established pedagogical traditions. [[Oates 2025 - ChatGPT-in-the-Classroom-Evaluating-Its|

Oates and Johnson (2025)]] demonstrate pedagogical value in making AI outputs “subjects for evaluation rather than endpoints of learning”—an approach that uses AI as material for critical analysis. [[Berry 2023 - AI-Ethics-and-Digital-Humanities|Berry (2023)]] warns that uncritical AI reliance risks “algorithmization” of Digital Humanities, where computational methods become defaults that shape questions before they are asked. ^teaching-and-pedagogy-p3

Service and Peer Review

Service and peer review use AI for triage, formatting checks, and detection tasks—identifying duplication, verifying references, smoothing language—while preserving human judgement for evaluations of originality, significance, and methodological rigor. [[Sun 2025 - Large-Language-Models-in-Peer-Review-Ch|Sun (2025)]] maps LLM applications in peer review, emphasizing that models “remain fundamentally limited in assessing research novelty”. [[Hosseini 2023 - Fighting-Reviewer-Fatigue-or-Amplifying|Hosseini and Horbach (2023)]] document potential to combat reviewer fatigue while identifying risks when “operational assistance crosses into epistemic territory,” suggesting that the boundary between support and substitution requires careful negotiation. [[Checco 2021 - AI-Assisted-Peer-Review|Checco et al. (2021)]] demonstrate AI can predict outcomes based on “superficial” features while emphasizing “semi-automated” approaches that preserve human judgement for consequential decisions. ^service-and-peer-review-p1

The line between operational assistance and epistemic authority runs directly through peer review, but it is better understood as a moving boundary than a fixed division. Tasks such as formatting checks, duplication detection, reference verification, and metadata checks may sit relatively comfortably on the operational side. Literature triage, reviewer matching, summary generation, and consistency checking occupy more contested middle ground. Assessments of novelty, significance, methodological rigor, interpretive adequacy, and contribution to a field require the kind of accountable judgement that remains distinctively human. Clarity about this boundary helps preserve the integrity of peer review while leveraging AI’s capacity for routine tasks. ^service-and-peer-review-p2

The inverse problem is integrity infrastructure. AI may assist legitimate review workflows, but it can also amplify older forms of scholarly fraud: paper mills, fabricated peer reviews, synthetic citations, citation cartels, manipulated images, and low-quality or fabricated manuscripts designed to pass superficial screening. These problems predate contemporary genAI, but generative systems increase their scale, speed, and plausible fluency. Responsible AI governance in publishing therefore requires attention not only to how reviewers may use AI, but also to how editorial systems detect, document, and respond to AI-amplified threats to the scholarly record. ^service-and-peer-review-p3

Research-Related Infrastructures

Research-related infrastructures span organizational governance, publishing and communication platforms, workflow systems, and accountability mechanisms. Governance frameworks, risk registers, roles, and capacity-building programs institutionalize stewardship, coordinating policy and practice across libraries, IT, research support, and ethics offices. [[Wu 2024 - AI-Governance-in-Higher-Education-Case|Wu, Zhang, and Carroll (2024)]] document Big Ten universities’ “multi-unit governance involving information technology, teaching centers, libraries, and research offices”—a distributed model recognizing AI as a cross-sectoral concern requiring coordinated response. ^research-related-infrastructures-p1

AI also places new pressure on open scholarly infrastructure. Repositories, commons platforms, digital research infrastructures, standards bodies, and open-science advising organizations must decide whether and how to integrate AI-mediated search, metadata generation, summarization, moderation, preservation, and analytics. These decisions affect governance, sustainability, procurement, interoperability, environmental cost, labour, and the degree to which open infrastructures remain accountable to scholarly and public communities rather than becoming dependent on proprietary AI services. ^research-related-infrastructures-p2

Open infrastructure responses should therefore be assessed against principles such as transparency, community governance, interoperability, portability, auditability, and long-term stewardship. In this context, standards for open infrastructure become AI governance instruments: they shape whether AI functions can be inspected, contested, replaced, or collectively maintained. ^research-related-infrastructures-p3

[[Papagiannidis 2025 - Responsible-Artificial-Intelligence-Gove|Papagiannidis, Mikalef, and Conboy (2025)]] differentiate governance practices, revealing tensions “between centralized oversight and distributed expertise.” Institutions must navigate between the efficiency of centralized decision-making and the contextual knowledge held by distributed practitioners who understand how AI affects their specific domains. ^research-related-infrastructures-p4

Rughiniș et al. (2025) introduce “dual black-boxing” capturing compound accountability challenges when opaque AI systems meet opaque institutional processes. When neither the AI nor the institution is transparent about its decision-making, accountability becomes doubly difficult. [[Janssen 2025 - Responsible-Governance-of-Generative-AI|Janssen (2025)]] reframes AI governance as managing “complex adaptive systems characterized by co-evolution and emergent properties,” suggesting that governance frameworks must themselves be adaptive. [[Weber 2023 - Organizational-Capabilities-for-AI-Imple|Weber et al. (2023)]]—whose findings on organizational capacity are discussed in the cross-cutting tensions above—appear here as a reminder that infrastructure governance is where those abstract tensions become concrete institutional failures. ^research-related-infrastructures-p5

Publishing and communication infrastructures increasingly integrate AI functions—journal recommendation, reviewer matching, manuscript screening, metadata generation—as part of scholarly workflow systems. [[Kousha 2024 - Artificial-Intelligence-to-Support-Publi|Kousha and Thelwall (2024)]] map AI tool deployment across the publishing pipeline, distinguishing demonstrated capabilities from promotional claims. Their analysis reveals that substantial efficiency improvements are achievable in labour-intensive administrative tasks, while human judgement remains integral to core intellectual evaluations. This distinction—between what AI can assist with operationally and what requires human epistemic judgement—structures responsible integration of AI into publishing infrastructures. Ultimately, however, the success of these infrastructural choices is measured not just by operational efficiency, but by how effectively they mediate knowledge for the diverse publics who rely on them. ^research-related-infrastructures-p6

Audiences and Differential Impacts[KD4]

Audiences—scholars, students, practitioners, policymakers, and broader publics—encounter AI-mediated summaries, translations, and recommendations that shape how they access and interpret scholarship. Each audience brings different needs, literacies, and vulnerabilities to these encounters, and AI effects manifest differently across these diverse contexts. ^audiences-and-differential-impactskd4-p1

Personalization raises the risk of filter bubbles, where recommendation systems surface content that confirms existing interests and perspectives while obscuring alternatives. [[Taneja 2020 - AI-Powered-Recommender-Systems-Personal|Taneja and Tripathi (2020)]] identify structural tensions inherent in personalization algorithms that enhance user experience while simultaneously constructing echo chambers that systematically exclude novel, dissenting, or interdisciplinary perspectives. For scholarly communication, filter bubbles could reinforce disciplinary silos and impede the cross-fertilization that drives innovation. Audience-aware communication emphasizes context, uncertainty, and reciprocity, recognizing that different audiences need different framings and levels of technical detail. ^audiences-and-differential-impactskd4-p2

The accessibility principles outlined earlier apply with particular force here: different audiences face different obstacles. AI can broaden participation and lower barriers—through translation, summarization, and alternative formats—but only when its design and governance prioritize inclusion, fidelity, and the preservation of diverse voices and perspectives. Da Silva Cardoso and Rocio (2025) propose frameworks for integrating AI into academic digital libraries that position librarians as essential partners in system development, emphasizing that understanding the nuanced and context-dependent needs of diverse academic communities requires human expertise that cannot be automated. The question is whether AI mediation grows or shrinks the publics that can engage with scholarship. ^audiences-and-differential-impactskd4-p3

The sources reveal how AI effects vary by context in ways that demand attention to differential impacts:

Institutional type. AI adoption is shaped by uneven institutional capacities, but these differences do not map neatly onto size or prestige. Large research universities may have more formal infrastructure, while also facing larger user bases, more complex bureaucracies, and less responsive or less humane interventions. Smaller institutions, community colleges, public libraries, independent scholarly organizations, and grassroots knowledge projects may face different constraints, including limited staffing, funding, or access to specialized expertise. Equitable AI adoption therefore requires governance models that can adapt across institutional contexts without assuming uniform access to technical capacity, administrative infrastructure, or training resources. ^audiences-and-differential-impactskd4-p4

Language communities. What serves English-language scholarship may disadvantage multilingual communities. Models trained predominantly on English reproduce Anglophone perspectives; multilingual support often lags behind. For scholars working in languages underrepresented in training corpora, AI tools may be less accurate, less helpful, or actively distorting. Linguistic diversity constitutes both an epistemic and an ethical concern: knowledge produced in non-dominant languages risks marginalization when AI systems favor majority languages. ^audiences-and-differential-impactskd4-p5

Career stage. What benefits established researchers may burden early-career scholars differently. Senior researchers with established reputations may more easily navigate AI-mediated environments, using AI assistance to amplify existing authority. Early-career scholars face additional pressures around attribution, originality, and demonstrating independent capability. When AI can generate fluent prose, summarize literatures, and accelerate administrative or writing tasks, how do junior scholars demonstrate the intellectual development that hiring, promotion and grant committees have traditionally valued? Assessment criteria developed for human-only production may require adaptation to remain meaningful in AI-augmented contexts, but such adaptation must avoid intensifying surveillance or creating new burdens of disclosure that fall unevenly on those with the least institutional power. ^audiences-and-differential-impactskd4-p6

AI in the evaluation of researchers deserves separate attention from AI in the peer review of scholarly outputs. Hiring, promotion, tenure, grants, fellowships, and institutional metrics shape careers, disciplines, and access to scholarly futures. If AI-mediated tools are used to screen CVs, rank candidates, summarize dossiers, evaluate productivity, or assess “fit,” their consequences may be especially significant for early-career scholars, contingent faculty, disabled scholars, scholars working in less dominant languages, interdisciplinary researchers, and those whose contributions are not well captured by standard metrics. Governance in this area should require transparency, contestability, human accountability, and careful attention to disparate impact. ^audiences-and-differential-impactskd4-p7

Disciplinary context. Different disciplines bring different norms, methods, and evaluation criteria to AI engagement. What constitutes appropriate AI assistance in a literature review differs from appropriate assistance in data analysis or creative work. Disciplinary communities are developing varied approaches, and cross-disciplinary dialogue helps surface assumptions and identify best practices that might transfer across contexts. ^audiences-and-differential-impactskd4-p8

Geographic location. Scholars in different regions face different infrastructural realities, regulatory frameworks, and access conditions. AI tools developed primarily for well-resourced contexts may not serve scholars working with limited bandwidth, different privacy regimes, or constrained institutional support. Geographic diversity in AI impacts connects to broader patterns of global inequality in knowledge production and circulation. ^audiences-and-differential-impactskd4-p9

Attention to these differential impacts marks responsible AI adoption, requiring ongoing assessment of who benefits, who bears burdens, and how distributions might be made more equitable. Rather than assuming uniform effects, responsible governance monitors for disparate impacts and adjusts policies and tools in response to evidence of inequitable outcomes. ^audiences-and-differential-impactskd4-p10

Phenomena, Relations, and Ongoing Inquiry

The sources in this annotated bibliography have been selected to trace several intersecting concerns. Conceptually, works that articulate critical frameworks (STS, data colonialism) offer key tools for understanding AI as a sociotechnical formation. ([[Anderson 2024 - AI-as-Philosophical-Ideology-A-Critical|Anderson 2024]]; [[Syed Mustafa Ali 2023 - Histories-of-artificial-intelligence-a|Ali et al. 2023]]; [[Bender 2021 - On-the-Dangers-of-Stochastic-Parrots-Ca|Bender et al. 2021]]). Normatively, scholarship on public engagement, participatory infrastructures, reciprocity, and accessibility grounds the discussion in the normative coordinates of open social scholarship ([[Arbuckle 2022 - An-Open-Social-Scholarship-Path-for-the|Arbuckle et al. 2022]]; [[El Khatib 2019 - Open-Social-Scholarship-Annotated-Biblio|El Khatib et al. 2019]]). Infrastructurally, sources on data governance (FAIR and CARE), licensing, repositories, and platforms illuminate how AI is being integrated into the material and organizational substrates of knowledge work ([[Carroll 2020 - The-CARE-Principles-for-Indigenous-Data|Carroll et al. 2020]]; [[White 2024 - The-Model-Openness-Framework-Promoting|White et al. 2024]]). Methodologically, contributions from anthropology, STS, media studies, information science, and environmental humanities help maintain epistemic plurality and attend to global and intersectional dimensions ([[Chung 2025 - Betting-on-UnCertain-Futures-Sociotec|Chung 2025]]; [[Richter 2025 - Imaginaries-of-Artificial-Intelligence|Richter, Katzenbach, and Schäfer 2025]]). Finally, empirically, studies of peer review, literature searching, pedagogical applications, and governance supply concrete evidence of AI's effects in practice ([[Sun 2025 - Large-Language-Models-in-Peer-Review-Ch|Sun 2025]]; [[Deng 2025 - Does-ChatGPT-Enhance-Student-Learning-A|Deng et al. 2025]]; [[Wu 2024 - AI-Governance-in-Higher-Education-Case|Wu, Zhang, and Carroll 2024]]). ^phenomena-relations-and-ongoing-inquiry-p1

Conceptual Mapping

The conceptual mapping that orients this bibliography treats concepts, phenomena, and relations as interconnected. This mapping is meant to help readers locate where a given source sits: what it names, what it observes, and what it connects: ^conceptual-mapping-p1

Concepts such as openness, provenance, bias, governance, data sovereignty, participation, reproducibility, labour, sustainability, platformization, multilingualism, accessibility, evaluation, enclosure, commons, personalization, and exposure diversity mark key terms of analysis. ^conceptual-mapping-p2

Phenomena such as pretraining on public corpora, proliferation of synthetic content, recommendation feedback loops, metadata drift, [[Hallucination|hallucination]], hybrid openness models, human-in-the-loop verification, green AI practices, community moderation, content credentials, and retrieval-augmented generation describe observable patterns in the field. ^conceptual-mapping-p3

Relations trace interactions among these elements: openness and enclosure meet through licensing and reciprocity mechanisms; provenance and trust are linked through audit trails and disclosure practices; bias and diversity intersect in dataset governance and evaluation design; governance and participation connect through co-design and community oversight; platformization shapes visibility through algorithmic curation and efforts to foster exposure diversity; labour and sustainability meet in discussions of ethical procurement and environmental accounting; reproducibility pushes against black-boxing through documentation and open benchmarking; multilingualism and accessibility intertwine in translation and community review; evaluation anchors legitimacy by attending to social impact and [[Equity|equity]] metrics. ^conceptual-mapping-p4

Focused Areas for Further Intervention

Several areas emerge from this scan as candidates for shorter, more pragmatic follow-up work. AI-mediated scholarly discovery is already reshaping canon formation, citation patterns and visibility through search engines, recommender systems, citation-context services, discovery platforms and automated synthesis tools. Provenance infrastructure, including content credentials, persistent identifiers, verifiable metadata, and workflow documentation, could strengthen disclosure by making AI involvement more portable, auditable and machine-readable. Smaller, community-built, purpose-specific models offer a constructive path for aligning AI with OSS values where communities can define scope, data governance, evaluation criteria, and benefit-sharing arrangements. ^focused-areas-for-further-intervention-p1

Integrity infrastructure also requires urgent attention, since paper mills, fabricated reviews, synthetic citations and AI-amplified fraud threaten the trust systems upon which scholarship depends. AI in the evaluation of researchers—hiring, promotion, grants, fellowships, and institutional metrics—deserves treatment distinct from peer review of outputs because its consequence for early-career scholars, equity and disciplinary futures may be especially profound. Parallel interventions on AI and open licensing, and on AI's effects on open scholarly infrastructure, would clarify how licensors, libraries, funders, digital research infrastructures, advising bodies and standards communities are adapting to machine use of open knowledge. ^focused-areas-for-further-intervention-p2

Conditions for Flourishing

The question animating this bibliography remains phenomenological: what conditions enable scholarship to flourish? The sources suggest these conditions include: ^conditions-for-flourishing-p1

- **Diverse participation** that brings varied perspectives to bear on complex problems
- **Critical literacy** that equips scholars and publics to evaluate AI-mediated claims
- **Transparent processes** that make choices visible and contestable
- **Human accountability** that maintains responsibility for consequential decisions
- **Infrastructural stewardship** that ensures technical systems serve public purposes

OSS principles provide the normative coordinates for navigating the cross-cutting tensions this scan has identified—offering a framework to ensure that operational assistance does not usurp epistemic authority, that openness resists extractive enclosure, that technical deployments match organizational capacity, and that efficiency does not erase the deliberative processes of scholarship. Specifically, openness with care balances access with consent and community control. Participation with shared authority embeds publics and communities in design and evaluation. Accessibility spans language, disability, and technical literacy. Reciprocity ensures that where community-sourced data or labour underwrite AI functions, benefits flow back to contributors. Infrastructural responsibility commits institutions to steward provenance, auditability, and accountability across deployment lifecycles. ^conditions-for-flourishing-p2

This mapping does not aim to close debate or fix categories. It sketches a topology of the current field: which nodes are active, where tensions gather, and how shifts in one domain reverberate through others. The annotated bibliography is offered as a baseline “lay of the land” against which ongoing research can be situated, helping scholars and practitioners locate their work within a broader conceptual ecology and identify gaps, alignments, and opportunities for intervention shaped by OSS principles of accessibility, reciprocity, and public value. ^conditions-for-flourishing-p3

These conditions also point towards a positive agenda. AI-mediated scholarship could support forms of access and engagement that have long mattered to OSS: translation across languages and registers, navigation of complex archives, discovery across disciplinary boundaries, support of disabled readers and writers, richer metadata, and new forms of dialogue between specialists and broader publics. Whether such uses weaken or strengthen scholarship depends less on the abstract presence of AI than on the infrastructures, governance processes, and values through which AI is developed and adopted. ^conditions-for-flourishing-p4

The landscape will continue to shift. New capabilities will emerge, governance frameworks will evolve, communities will develop practices and norms through experimentation and deliberation. What this bibliography offers is a documented moment from which change can be measured and assessed. The sources assembled here provide conceptual resources for that ongoing work: frameworks for analysis, evidence of effects, models for governance, and principles for evaluation. The task ahead is to use these resources thoughtfully, maintaining the commitments to accessibility, reciprocity, and public engagement that make scholarship worth pursuing. ^conditions-for-flourishing-p5